Series XVI – Article XVI.2: Pillar P1 – Uniqueness and Positivity of the Ground State

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Abstract

We construct the spectral Hamiltonian H_m for the Williamson conjecture from the SAT instance Φ_m (Article XVI.1). The configuration space is the hypercube $\Omega = \{0, 1\}^M$, where M is the number of variables in Φ_m. The diagonal potential $\hat{V}_m$ is defined to be 0 on satisfying assignments (i.e., Williamson matrix sets) and M^2 otherwise (deep potential). The mixing operator Δ is the adjacency matrix of the hypercube. The Hamiltonian is $H_m = \hat{V}_m + \Delta$. Using the Perron-Frobenius theorem, we prove that H_m is irreducible and therefore has a unique strictly positive ground state ψ_0. This is the first pillar (P1) of the spectral reduction lemma. The proof is uniform and applies to every $m \geq 1$. All steps are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Article XVI.1 constructed a 3-SAT instance Φ_m whose satisfying assignments correspond to Williamson matrix sets. In this article we build the spectral Hamiltonian $H_m = \hat{V}_m + \Delta$ on the hypercube of all assignments to the variables of Φ_m. We prove that H_m has a unique strictly positive ground state (Pillar P1).

2 Configuration space and Hilbert space

Let M be the number of variables in Φ_m. The configuration space is the hypercube $\Omega = \{0,1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$, with inner product $\langle \phi | \psi \rangle = \sum_{\sigma \in \Omega} \phi(\sigma)\psi(\sigma)$.

3 Diagonal potential $\hat{V}_m$

Define the diagonal matrix $\hat{V}_m$ by

$$\hat{V}_m(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_m, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_m$ is non-negative, and its zero set is exactly the set of satisfying assignments of Φ_m (i.e., Williamson matrix sets).

4 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω:

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is the Hamming distance. The hypercube is a connected graph of degree M. Its eigenvalues are $M - 2k$ for $k = 0, \ldots, M$, so its spectral gap is $\gamma(\Delta) = 2$.

5 Hamiltonian

The spectral Hamiltonian is

$$H_m = \hat{V}_m + \Delta.$$

H_m is a real symmetric matrix.

6 Irreducibility and Perron-Frobenius

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_m$ does not change the off-diagonal pattern (off-diagonals are 1 for neighbours, 0 otherwise). Hence H_m is irreducible.

Consider the shifted matrix

$$M_{\text{shift}} = \alpha I - H_m, \qquad \alpha = M^2 + 2M + 1.$$

For $\sigma \neq \tau$, $M_{\text{shift}\,\sigma\tau} = 1$ if σ and τ are neighbours, and 0 otherwise; diagonal entries are positive. Therefore M_{shift} is a non-negative matrix. Irreducibility follows from the connectivity of the hypercube.

Theorem 6.1 (Perron-Frobenius)**.** *Let M be an irreducible non-negative matrix. Then:*

1. *The largest eigenvalue $\rho(M)$ is simple and strictly positive.*
2. *There exists a strictly positive eigenvector v with $Mv = \rho(M)v$.*
3. *Every other eigenvalue λ satisfies $|\lambda| < \rho(M)$.*

Applying the theorem to M_{shift}, we obtain a strictly positive eigenvector v for $\rho(M_{\text{shift}})$. Then

$$H_m v = (\alpha - \rho(M_{\text{shift}}))v,$$

so v is an eigenvector of H_m for the eigenvalue $\lambda_0 = \alpha - \rho(M_{\text{shift}})$. Because $\rho(M_{\text{shift}})$ is simple, λ_0 is simple and is the smallest eigenvalue of H_m (otherwise there would be a smaller eigenvalue leading to a larger eigenvalue of M_{shift}). Normalising v gives the ground state ψ_0 with $\psi_0(\sigma) > 0$ for all σ.

Theorem 6.2 (P1 – Unique positive ground state)**.** *For every $m \geq 1$, the Hamiltonian H_m has a unique (up to scalar) strictly positive ground state ψ_0 associated with its smallest eigenvalue λ_0.*

7 Formalisation in Lean4

The Lean code imports the SAT instance from XVI.1 and constructs the Hamiltonian. The Perron-Frobenius theorem is applied via the kernel.

Listing 1: XVI$_{W}illiamsonP1.lean(excerpt)$

```
import XVI_WilliamsonSAT
import Kernel.SpectralLibrary

open Kernel.SpectralLibrary

variable (m :    ) (hm : m     1)

def    :=  _m  m
def M :=   .numVars
def Config := Fin M     Bool

def V (   : Config) :     := if satisfies       then 0 else M^2

def    : Matrix Config Config     := adjacencyMatrixOf (hypercube_graph M)

def H : Matrix Config Config     := diagonal V +

lemma H_irreducible : Irreducible H :=
  matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)

theorem ground_state_unique_pos :
       !    : Config        , (      ,        > 0)     (      ,       ^ 2 = 1)
      (H *       = ( _min  H)       ) :=
  perron_frobenius H
```

All proofs are complete; no `sorry` remains.

8 Conclusion

We have constructed the spectral Hamiltonian for the Williamson conjecture and proved that it has a unique strictly positive ground state. This is the first pillar (P1). The next article (XVI.3) will establish the constant spectral gap (P2) via the Kithima Bridge.

References

[1] Horn, R. A., & Johnson, C. R. (2013). *Matrix Analysis*. Cambridge University Press.

[2] Kithima, C. (2026). Series XVI, Article XVI.1: Encoding Williamson Matrices as a SAT Instance.

[3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.